How uncertain is a posterior-predictive 1-in-100-year level?

A synthetic-data check of the Müller–Watson Bayesian model

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https://github.com/Sustainable-Solutions-Lab/extreme-trend-detection

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Summary

The question. A 1-in-100-year level can be read off a Bayesian model of time-varying extremes in two ways. One takes a statistic (median or mean) of the 99th percentiles implied by the individual posterior draws. The other, which we call *the predictive level*, first averages the draws' distributions into one posterior predictive distribution and then takes its 99th percentile. The predictive level is a single, well-defined number, and it is the one a learner who holds the posterior should use to price a contract that pays when the level is exceeded. We asked how close that number is to the truth, and where its uncertainty comes from, using synthetic records whose true risk is known exactly.

What we did. We fitted the four-random-walk GEV model of Müller and Watson (2024), with the authors' code and priors, to synthetic records of annual maxima (30, 45 or 85 years) drawn from four daily distributions (19,700 model fits), including a precipitation-like gamma distribution, with and without known changes in their mean, spread or tail. In each record we computed the predictive 99th percentile for the final year and asked how often that level is truly exceeded.

On average across records the predictive level is about right or conservative. The true chance of exceeding it averages 0.85% at 85 years and 0.4–0.6% at 30–45 years, against the 1% it is meant to have, whereas the posterior median and the maximum-likelihood fits average 1.6–3.7%. It rarely understates the risk (13% of records at 85 years, 6–8% at 30–45 years, against 30–40% for the other estimators).

In a single record it usually overstates the risk, and often by a lot. The median true chance of exceeding it is 0.39% at 85 years, 0.05% at 45 years and 0.01% at 30 years: the level reported as the 1-in-100-year event is typically a 1-in-250, 1-in-2,000 and 1-in-10,000-year event. It is off by more than a factor of 2 in either direction in 68%, 84% and 92% of records, 9 to 19 percentage points more often than the posterior median or a simple trend fit. In the precipitation-like case, the predictive 100-year depth is typically 8% (85 years), 22% (45) and 31% (30) above the true one, and in one record in ten more than 25%, 66% and 138% above.

As a price, it is conservative and very variable. For cover whose true risk is 1% a year, the model's predictive probability, its pure premium rate, has a median of 1.7% at 85 years and 2.8–3.2% at 30–45 years. It is rarely below 0.5% (4–10% of records), whereas a location-trend fit underprices by more than a factor of 2 in 36–45% of records and by more than a factor of 10 in 16–28%.

The uncertainty comes from the short record and the prior, not from the computation.

- Across equally plausible records of the same climate, the true chance of exceeding the predictive level spans a factor of about 160 at 85 years (10th to 90th percentile 0.015% to 2.4%), and far more at 30–45 years.
- Within a record, the posterior itself is very uncertain about the risk at that level: the typical

draw puts it at 0.2% (85 years) or essentially 0% (30 years), while the top 5% of draws put it above 4.6–5.8%. The 1% is an average over beliefs that range from almost nothing to five or more times the target. The top 5% of draws carry 38–52% of it.

- The overstatement is largest in records whose posterior leans towards heavy tails (a positive GEV shape ξ), which none of our synthetic climates had in the final year.
- Shrinking the prior scales of the random walks (a "tight prior") lowers the predictive level, raises its typical true exceedance from 0.39% to 0.64% at 85 years and from 0.05% to 0.24% at 45 years, and brings its accuracy close to that of the posterior median.
- Monte Carlo error is negligible: with 4,000 draws it changes the exceedance probability by about 5–10%, and with the authors' 40,000 draws by 2–3%; it is well under 1% of the error variance.

Most of the ambiguity is there even when nothing is changing. In records from a stationary climate the predictive level is just as dispersed and just as high: it misses by more than a factor of 2 in 64%, 80% and 89% of records at 85, 45 and 30 years. The high and volatile risk estimates come mainly from estimating a heavy upper tail from a few decades of data under a prior that allows very heavy tails, not from non-stationarity as such.

What we would suggest. Report, beside the predictive level, how the posterior's beliefs about the risk at that level are spread (e.g. the median draw and a 90% range); show how the level moves under a tighter prior on the random walks and on ξ; fit the same model to stationary synthetic records of the same length to see how much ambiguity arises with no change at all; and use pooled regional information on the tail shape where possible (Section 7).

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1 What we tested

1.1 The predictive level

The model gives, for each posterior draw $d = 1, \ldots, D$, GEV parameters $\theta_d(t) = (\mu_d(t), \sigma_d(t), \xi_d(t))$ for the annual maximum in year t. The draw's own 99th percentile is $q_d(t)$, with $P(Y_t > q_d(t) \mid \theta_d) = 1\%$. The *posterior median* estimate is the median of the $q_d(t)$ over draws. The *predictive level* $x^*(t)$ instead solves

$$\frac{1}{D}\sum_{d=1}^{D} P(Y_t > x^*(t) \mid \theta_d(t)) = 1\%, \tag{1}$$

the 99th percentile of the posterior predictive distribution of the year-t maximum. It averages probabilities, not levels. Since a probability cannot go above 1 but can fall to zero, and the upper tail of a GEV distribution is very sensitive to ξ, the average is pulled up by the draws with the heaviest tails: in our records x^* is above the posterior median in every record, and above the posterior mean of q_d in two-thirds of records at 85 years (a tenth at 30 years; Figure 3).

We assume that the reported "99th percentile of the posterior rainfall distribution" is this predictive 99th percentile of the annual-maximum distribution, i.e. the 1-in-100-year level, in the final year of the record. If it is a percentile of another quantity (daily rainfall, say), the mechanism below still applies but the numbers will differ.

1.2 The model, the data and the score

Model The four-random-walk GEV model of Müller and Watson (2024), their Stan code and priors unchanged (random walks in ξ, a scale and two location terms; innovation scales $\gamma = \mathrm{sg}\cdot g/\sqrt{T}$ with $g \sim \mathrm{Exp}(1)$ and $\mathrm{sg} = 1, 0.5, 0.5$; ξ confined to $(-0.9, 2.1)$). Four chains of 1,000 warm-up and 1,000 sampling iterations (`adapt_delta` 0.95). The "tight prior" runs use $\mathrm{sg} = 0.25$ for all three.

Data Daily series with day-to-day correlation 0.5 (0 in one design) from four distributions: normal (temperature-like), gamma with shape 2 (precipitation-like), Student-t with 6 degrees of freedom (heavy-tailed) and a bounded beta. We fitted the annual maxima (or the three largest values per year, declustered, in the "3/yr" designs). Each distribution was left unchanged or had its mean, spread, tail or (beta) position changed linearly or in a step; further cases make the spread and tail follow a rising mean, as a covariate model would assume, or change linearly in time between the same end points. Each case has 100 records.

Truth Because the generating process is known, so is the true distribution of the final year's annual maximum, and with it the true chance of exceeding any reported level.

Score A reported 1-in-100-year level *understates* the risk by more than a factor of 2 if its true chance of being exceeded is above 2%, and *overstates* it if that chance is below 0.5%. We call the sum the share *missed*.

Comparison The same records were scored with the posterior median and mean of q_d, a maximum-likelihood GEV fit with a linear trend in location, and an Akaike-weighted average of four simple fits (constant, location trend, location and width trend, mean-state covariate). Because all estimators see the same records, differences between them are measured to about ±1.5 percentage points.

2 How close is the predictive level to the truth?

Table 1 and Figure 1 give the main result, pooled over the standard cases (the four distributions with and without change).

Table 1: True chance of exceeding the reported 1-in-100-year level in the final year, standard cases (2,000 records per design). For the predictive level: median and average over records, and the shares (%) understated ($> 2\%$) and overstated ($< 0.5\%$) by more than a factor of 2. Right: the share missed by the other estimators on the same records, and the difference predictive minus posterior median (± one standard error).

	Bayes, predictive					Missed, others			
Design	median	avg.	under	over	**missed**	post. median	loc. trend	AIC avg.	pred. − median
30 yr	0.01%	0.5%	8	84	**92**	75	73	75	$+17 \pm 1$
45 yr	0.05%	0.4%	6	78	**84**	70	69	69	$+14 \pm 1$
85 yr	0.4%	0.8%	12	56	**68**	59	56	51	$+9 \pm 2$
85 yr, 3/yr	0.5%	0.9%	13	53	**66**	55	55	46	$+11 \pm 2$
85 yr, indep. days	0.3%	0.8%	13	58	**71**	65	58	55	$+6 \pm 2$
45 yr, tight prior	0.2%	0.7%	11	63	**74**	70	69	69	$+4 \pm 1$
85 yr, tight prior	0.6%	1.1%	16	42	**57**	56	56	51	$+1 \pm 1$

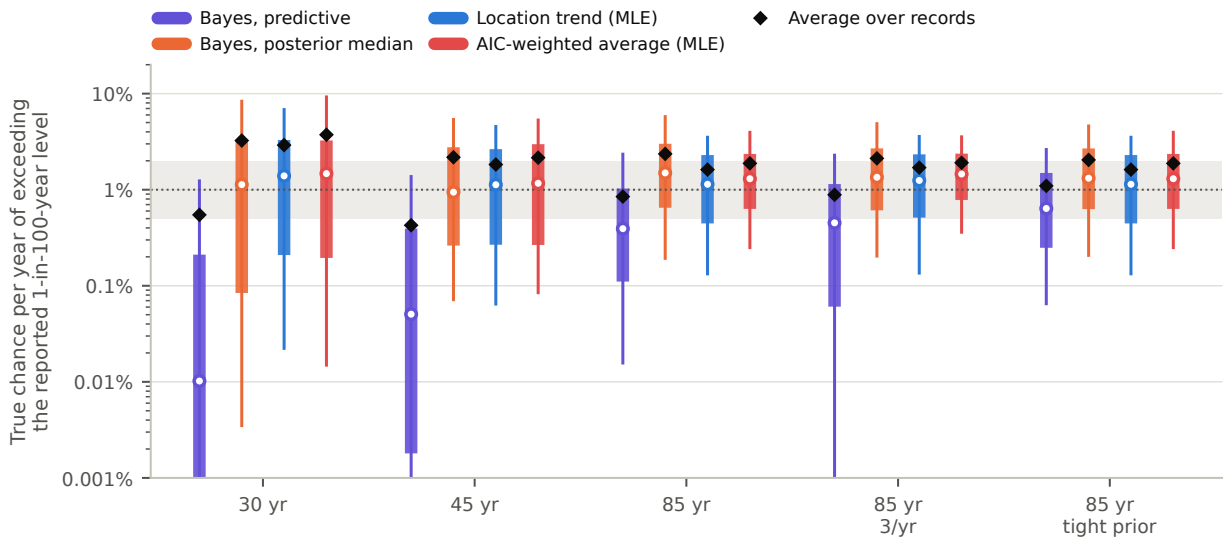


Figure 1: True chance of exceeding the reported 1-in-100-year level, over the records of the standard cases: 10–90th percentiles (thin bar), 25–75th (thick bar), median (circle) and average (diamond). The grey band is the factor-of-2 band around 1%.

- **The predictive level is usually too high.** Its median true exceedance is 0.39% at 85 years, 0.05% at 45 and 0.01% at 30. The other estimators are centred near 1% (medians 0.9–1.5%).
- **It misses more often.** It is off by more than a factor of 2 in 68% of records at 85 years, 84% at 45 and 92% at 30, almost always by overstating. The posterior median misses in 59%, 70% and 75%, the location trend in 56%, 69% and 73%.
- **The dispersion is larger, and lopsided.** At 85 years the middle 80% of its true exceedances runs from 0.015% to 2.4%, a factor of about 160; for the posterior median it is 0.19% to 6%, a factor of about 30. At 30 years, a tenth of predictive levels have essentially no chance of being exceeded in the simulated climate.
- **The level itself is not far off; the probability is.** In the precipitation-like (gamma) records, the predictive 100-year depth is a median 8% above the true depth at 85 years (10–90th percentile −6% to +25%), 22% above at 45 years and 31% above at 30. Because the chance of exceedance falls steeply with depth in the far tail, an 8% overstatement of the depth becomes a factor of 2.5 in the probability.

Table 2 breaks the 85-year results down by the kind of change and by distribution. The predictive level overstates the risk in every kind of case, including no change at all. It does comparatively well only where the true change makes the upper tail heavier, so that the other estimators understate the risk: when the spread and tail follow a rising mean, it misses in 68% of records, about the same as the posterior median (69%).

Table 2: The predictive level at 85 years by kind of change and by distribution: median true exceedance, shares (%) understated and overstated by more than a factor of 2, and missed; right, the share missed by other estimators on the same records.

		Bayes, predictive				Missed, others		
Kind of change	records	median	under	over	**missed**	post. median	loc. trend	AIC avg.
No change	400	0.5%	11	52	**64**	53	49	47
Mean	800	0.4%	14	54	**67**	57	52	45
Width	400	0.6%	19	46	**64**	63	54	54
Tail	300	0.2%	3	76	**79**	60	75	62
Position (beta)	100	0.2%	11	66	**77**	76	73	61
Follows mean state	700	0.6%	24	45	**68**	69	74	64
Time-driven	300	0.6%	24	47	**71**	68	71	65
Normal (standard cases)	400	0.5%	15	50	**65**	60	50	46
Gamma (standard cases)	500	0.4%	11	56	**67**	53	50	47
Student-t (standard cases)	500	0.4%	9	55	**64**	50	56	46
Beta (standard cases)	600	0.3%	15	60	**75**	70	66	60

3 The average versus a single record

The predictive level does what it is built to do on average: the true chance of exceeding it, averaged over records, is 0.85% at 85 years (0.43–0.55% at 30–45 years), the closest to 1% of the estimators we tested, alongside the posterior mean; the posterior median and the maximum-likelihood fits average 1.6–3.7% (diamonds in Figure 1). But the average is carried by a minority of records in which the level is too low, while in the typical record it is far too high.

This is the familiar gap between the average and the typical value of a skewed quantity. For a risk taker who writes many independent contracts and cares only about the expected total, the average is the relevant number, and by that standard the predictive level is well calibrated or conservative. For a single contract, location or portfolio exposed to one climate, the typical record is more relevant, and there the predictive level usually implies a risk several times higher than the true one. Climate risks at nearby locations are strongly correlated, and a model and prior applied in the same way everywhere will err in the same direction everywhere, so the diversification that would make the average relevant is limited.

A Bayesian predictive distribution is calibrated in this average sense when the true climates are drawn from the prior. Our synthetic climates are not drawn from the authors' prior, and in particular none of them has a heavy upper tail in the GEV sense in the final year; the prior allows ξ up to 2.1, where even the mean of the distribution is infinite. To the extent that real rainfall extremes have lighter tails than the prior allows, the same overstatement should be expected in real records.

4 Pricing cover with a known 1% risk

Figure 2 turns the question around: for the level whose true chance of being exceeded is exactly 1% in the final year, what chance does each model assign? That chance is the pure premium rate

the model would charge for cover triggered at that level.

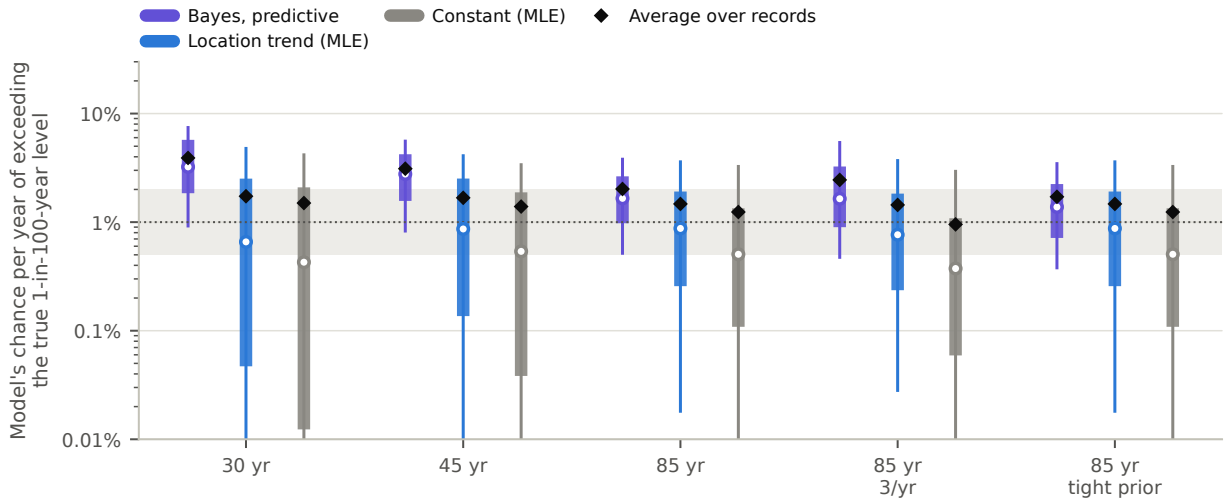


Figure 2: Each model's chance of exceeding the true 1-in-100-year level of the final year, over the records of the standard cases (percentiles as in Figure 1). The grey band is a factor of 2 around the true 1%.

- **The predictive model charges too much, but rarely too little.** Its median rate is 1.7% at 85 years, 2.8% at 45 and 3.2% at 30. It charges less than 0.5% in only 4–10% of records and less than 0.1% almost never; it charges more than 2% in 40% of records at 85 years and in 67–72% at 30–45 years.
- **A trend fit is right on average but often dangerously low.** The location-trend maximum-likelihood fit has a median rate of 0.7–0.9%, but charges less than 0.5% in 36–45% of records and less than 0.1% in 16–28%: in those records the cover would be priced at a tenth or less of its true cost.
- The predictive approach therefore trades frequent and large over-pricing for protection against severe under-pricing. Whether that is a good trade is a question about risk preferences; the point here is that the loading it implies is large, varies by a factor of several from one record to the next, and is set mainly by the prior (Section 5.3).

5 Where the uncertainty comes from

5.1 The posterior's own spread

Figure 3 shows one typical precipitation-like record of 85 years with no change. Left: each grey curve is the chance of exceeding a given level under one posterior draw. The draws fan out in the tail. The predictive curve (violet) is their average; it lies above most of the draws because the few heavy-tailed ones dominate the average in the far tail. Right: the distribution over draws of the chance of exceeding the predictive level. By construction it averages 1%, but the median draw gives 0.40%, close to the truth (0.48%), and the top 5% of draws carry a third of the average.

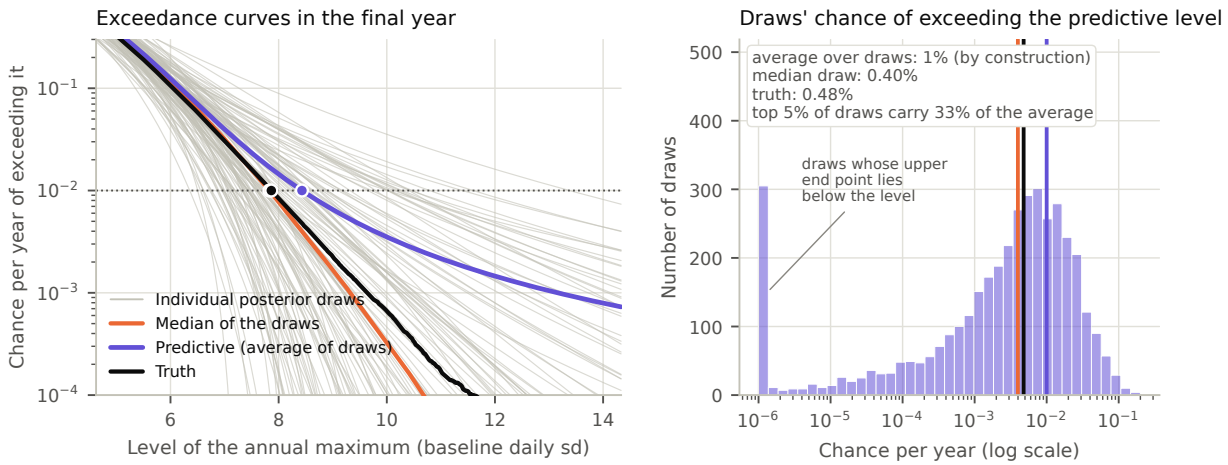


Figure 3: One record (precipitation-like distribution, 85 years, no change; the record whose predictive level has the median true exceedance of that case). Left: exceedance curves of 120 posterior draws (grey), their median (orange), their average, i.e. the predictive curve (violet), and the truth (black); dots mark the levels with a 1% chance. Right: the draws' chance of exceeding the predictive level (log scale; draws that put it at or below 10^{-6} are gathered in the leftmost bar).

Table 3 summarises this over all records. The posterior is honest about its uncertainty: the true chance lies within the draws' 5–95% range in 96–100% of records. But the range is very wide, from essentially 0 to about 5%, so the single number hides a very wide spread of beliefs. "Dominated by a few draws" is accurate in a specific sense: the top 5% of draws carry 38% (85 years) to 52% (30 years) of the 1%, and the effective number of draws that determine the average is 13–21% of the total.

Table 3: Uncertainty measures of the predictive level, standard cases. Left: the median over records of the 5th, 50th and 95th percentiles over draws of each draw's chance of exceeding the predictive level. *Heavy-tailed*: the share of records in which more than 5% of the draws have $\xi > 0.5$. *Truth inside*: how often the true chance lies within the draws' 25–75% and 5–95% ranges. *Top 5%*: the share of the 1% carried by the top 5% of draws (median over records). *Monte Carlo*: the typical error of the exceedance probability with 4,000 draws, as a multiplicative factor.

	Draws' chance of exceeding the level				Truth inside			
Design	5%	median	95%	heavy-tailed	50%	90%	top 5%	Monte Carlo
30 yr	0.00%	0.00%	5.8%	53%	64%	96%	52%	×1.11
45 yr	0.00%	0.04%	5.4%	38%	69%	100%	47%	×1.10
85 yr	0.00%	0.2%	4.6%	12%	67%	98%	38%	×1.06
85 yr, 3/yr	0.00%	0.3%	4.0%	9%	69%	97%	29%	×1.05
85 yr, indep. days	0.00%	0.2%	4.6%	17%	61%	98%	38%	×1.06
45 yr, tight prior	0.00%	0.3%	4.4%	18%	54%	97%	34%	×1.06
85 yr, tight prior	0.00%	0.5%	3.8%	4%	57%	96%	28%	×1.03

5.2 The tail shape

Which records are overstated most? Those whose posterior leans towards heavy tails (Figure 4). The true GEV shape of the annual maxima in our climates ranges from about −0.25 (bounded) to about +0.17 (Student-t); with 30–45 years of data the posterior mean of ξ is often well above that, and every positive shift in ξ raises the predictive level steeply. At 30 years, records with a posterior mean ξ above about 0.2 have predictive levels with essentially no chance of being exceeded.

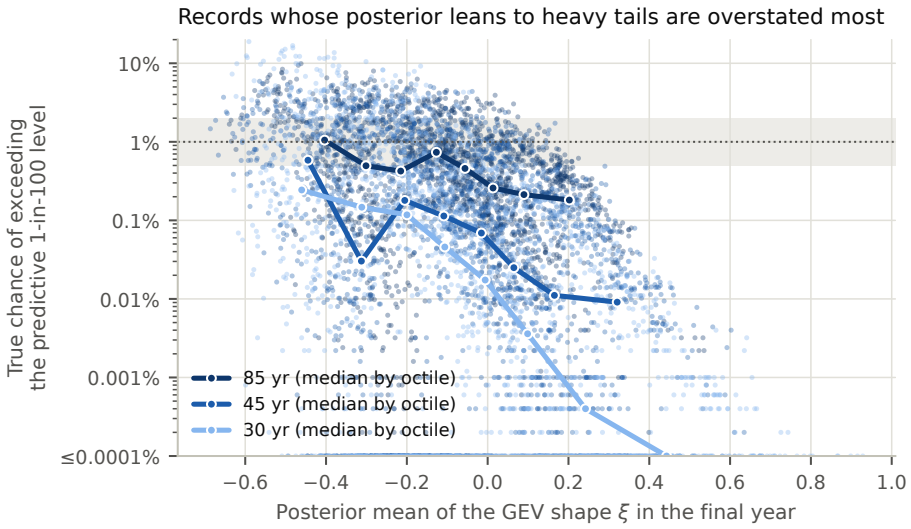


Figure 4: True chance of exceeding the predictive level against the posterior mean of the GEV shape ξ in the final year, standard cases at 30, 45 and 85 years (one dot per record; lines: medians within octiles of ξ).

5.3 The prior

The predictive level is "unique" only for a given prior, and the prior matters a great deal. Shrinking the prior scales of the random walks from the authors' values (1, 0.5, 0.5) to 0.25 changes nothing about the data, but:

- at 85 years the share missed falls from 68% to 57% (now within 1 point of the posterior median and the location trend), and the median true exceedance rises from 0.39% to 0.64%;
- at 45 years the share missed falls from 84% to 74% and the median true exceedance rises from 0.05% to 0.24%; record by record the true exceedance of the predictive level rises by a median factor of 2.8 (85 years: 1.6);
- in the cases where the spread and tail follow a rising mean, at 85 years, the predictive level with the tight prior becomes the best estimator we tested, missing in 55% of records against 64% for the Akaike-weighted average and 69% for the posterior median.

We varied only the random-walk scales. The prior on ξ itself, and its upper limit of 2.1, are likely to matter as much, because the overstatement comes from heavy-tailed draws.

5.4 Monte Carlo error

Splitting each fit's 4,000 draws into its four chains, the predictive level's chance of being exceeded changes by a typical factor of 1.06 at 85 years and 1.10–1.11 at 30–45 years (90th percentile 1.16–1.55). With the authors' 40,000 draws these factors would shrink by about $\sqrt{10}$. Monte Carlo error accounts for well under 1% of the variance of the error in the log exceedance probability (0.03–0.11%): the dominance of a few draws is a property of the posterior, not of the number of draws.

6 Ambiguity in the climate, or in the inference?

The argument for the predictive approach is that possible non-stationarity, combined with possibly long tails, makes the climate distribution genuinely ambiguous, and that the high and volatile estimates of catastrophic risk reflect that ambiguity. The synthetic records let us separate the two parts, because in the no-change cases nothing is changing and the true tail is never heavy:

Table 4: No-change cases only (400 records per design): true chance of exceeding the reported 1-in-100-year level (median and 10–90th percentiles) and the share missed by more than a factor of 2.

	Bayes, predictive			Missed > 2×		
Design	median	10%	90%	predictive	post. median	loc. trend
30 yr	0.02%	0.00%	1.35%	89	73	73
45 yr	0.08%	0.00%	1.46%	80	65	66
85 yr	0.47%	0.04%	2.31%	64	53	49
85 yr, tight prior	0.68%	0.12%	2.30%	52	48	49

With no change at all, the predictive level is as high and as dispersed as in the cases with change. So most of the ambiguity is produced by the inference: a few decades of annual maxima cannot rule out a heavy tail, the prior allows one, and averaging probabilities over the posterior turns that possibility into a high level. That is a legitimate feature of a Bayesian learner's beliefs, and it is real ambiguity for someone who must act on a short record. But it would be present in a stationary climate, it shrinks with record length roughly as the tail becomes identifiable, and its size is governed by prior choices that the data cannot check. It is not, by itself, evidence that the climate's extremes are changing or are heavy-tailed.

7 Suggestions for practice

1. **Report the spread of beliefs, not only the average.** Alongside the predictive level, report the posterior distribution of the chance of exceeding it, for example the median draw and a 90% range (in our records typically ∼0.2% and 0–5% at 85 years). That makes clear that "1%" is an average over beliefs ranging from negligible to several times the target.

2. **Show prior sensitivity.** Refit with tighter random-walk scales and a more informative prior on ξ (or a lower upper bound), and report how the level and the implied premium move. In

our records, the random-walk scales alone change the typical true exceedance by a factor of 1.6 (85 years) to 2.8 (45 years).

3. **Benchmark against a stationary climate.** Fit the model to synthetic stationary records of the same length and similar distribution. The spread of predictive levels obtained there is the ambiguity that comes from the inference alone.

4. **Compare with simple fits.** If the predictive level is far above both a constant and a trend maximum-likelihood fit, the difference comes from the posterior's heavy-tailed draws, and should be justified on other grounds (physical or regional knowledge of the tail).

5. **Bring in more information on the tail.** Pooling the shape parameter across nearby stations (regional frequency analysis), or using several values per year, constrains ξ much better than a single annual-maximum record, and the overstatement shrinks as ξ is pinned down.

6. **Separate the risk margin from the estimate.** The predictive approach builds a loading for ambiguity into the estimate through the prior. If a margin for ambiguity is wanted, it may be clearer to estimate the risk with a well-calibrated central estimate and add an explicit, stated margin, so that the margin is a decision rather than a by-product of the prior.

8 Caveats

- Synthetic climates: four distributions and a limited set of changes. Real rainfall extremes may have heavier tails than ours; if their true ξ is clearly positive, the predictive level will overstate less, and the other estimators will understate more.

- One model, with the authors' priors and one alternative for the random-walk scales. We did not vary the prior on ξ.

- We scored the final year of each record (a "nowcast"); a forecast for future years would add the uncertainty of extrapolating the random walks.

- 4,000 draws per fit rather than the authors' 40,000 (shown above to matter little), and, as in the authors' replication code, no importance-sampling correction for the extended-support likelihood (whose weights they report to be essentially 1).

- 100 records per case. Pooled shares are measured to about ± 1 percentage point, and differences between estimators on the same records to about ± 1.5.

A Results at 30 years by kind of change

Table 5: As Table 2, at 30 years.

		Bayes, predictive				Missed, others		
Kind of change	records	median	under	over	**missed**	post. median	loc. trend	AIC avg.
No change	400	0.02%	7	82	**89**	73	73	74
Mean	800	0.01%	9	84	**92**	75	69	72
Width	400	0.02%	10	81	**91**	77	76	79
Tail	300	0.01%	3	92	**95**	73	79	74
Position (beta)	100	0.00%	7	88	**95**	80	76	77
Follows mean state	700	0.01%	9	78	**87**	80	83	82
Time-driven	300	0.01%	9	81	**90**	78	83	81
Normal (standard cases)	400	0.01%	8	82	**90**	75	72	76
Gamma (standard cases)	500	0.02%	7	84	**91**	72	70	73
Student-t (standard cases)	500	0.04%	7	84	**91**	73	70	69
Beta (standard cases)	600	0.00%	8	87	**95**	79	79	80

B Reproducing these results

The Bayesian fits are rerun with `python scripts/stage_c.py -replay`, which refits every saved scenario with its original seeds and settings and adds the predictive quantiles, the per-chain values, the spread of the draws' exceedance probabilities and the final-year posterior draws (written to `data/output/stage_c_pred`; about 20 hours on 14 cores). The posterior medians and means reproduce the earlier runs to within 3×10^{-6}. The tables, figures and numbers of this note come from `python scripts/predictive_note.py`, which also writes the summary tables `predictive_headline.csv`, `predictive_uncertainty.csv` and `predictive_rain_ratio.csv` in `data/results`.